Dandelion: Hundreds of Millions of Distributed Replicated Transactions with Few Machines

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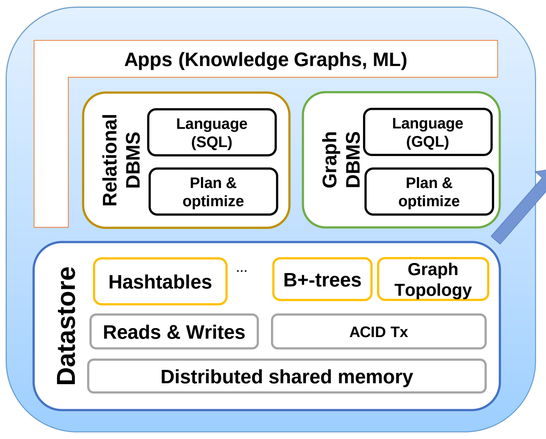
Datastores 101



Backbone of wide range of applications, services and systems (including DBMS)

Performance-critical intra-datacenter storage layer to access & manipulate data with

> *Efficiency*
high throughput, cost-effective

> *Programmability*
transactions, strong consistency

> *High availability*
replication and fault tolerance

Evolution of Datastores
(note: 10x leaps that need redesign)

Traditional Era

Traditional Datastores
Disk-based, slow network
➢ Efficiency: non-scalable, low performance

#1

NoSQL Datastores
Sacrifice programmability for performance
Ok for best-effort web-service but not critical app!
➢ Programmability: sacrifice transactions, consistency

#2
Exploit new HW for efficiency + programmability

New Era

Modern Datastores: e.g., Microsoft's FaRM [SOSP'15]
➢ Scalable, good perf., transactions, consistency, availability
- In-memory
- Fast (RDMA) network
- *FaRM* ➔ **140M TATP txs/sec (90 machines – 5TB)**

#3

Next-gen Datastores: academia + our goal
➢ Modern datastores + better perf., less cost, higher capacity
- HW acceleration: FPGA, ASICs, smart NICs/switches
- CXL expansion, Disaggregated memory, PMEM

More memory with less compute ➔ 50TB in 5-10 machines

Challenge: Can we achieve same or better performance with 10x fewer machines (compute) ?

Stoppers!

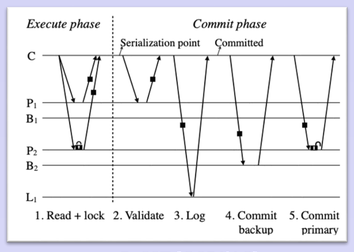
1. Fast **Networks**, but small packets ✗

***eRPC* [*NSDI'19 – best paper*]**
Efficient general-purpose RPCs
DPDK, UDP, RoCE/IB (two-sided) RDMA
Retransmissions
Support packets > MTU
- General purpose, Reliable, callback API
- Great for large/medium messages … **BUT**

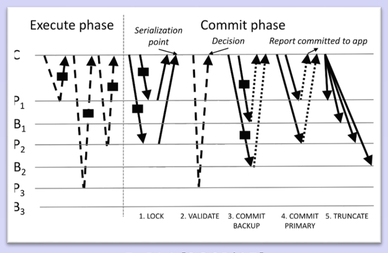
Issues with small (8-24B) requests
1. RoCE packets have 66B Header = 10-30% BW utilization
2. Too many packets, NIC/switch cannot keep up with packet rate

Datastores are dominated by
Small data messages → access/update small key-value pairs (e.g. 8-32B)
Tiny protocol logic messages → lock/unlock, commit/abort, truncate, etc.

2. Fault-tolerant strongly-consistent **Protocols** ✗
but complex and costly execution/commit phases
- lock-based execution phase or
- complex multiple round-trip (RTT) commit
- protocols have tradeoffs, but datastores no protocol alternatives

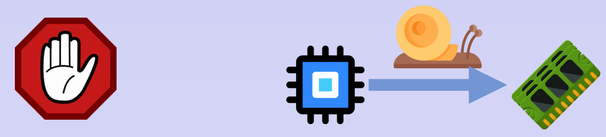


FaSST [OSDI'16]



FaRM [SOSP'15]

3. Concurrent in-memory **Data Structures** ✗
but practically blocking & not memory-aware



4. **CPUs** with many cores ✗
but cycles are wasted
on small packets, many protocol actions,
stalled waiting on memory, one-at-a-time request execution

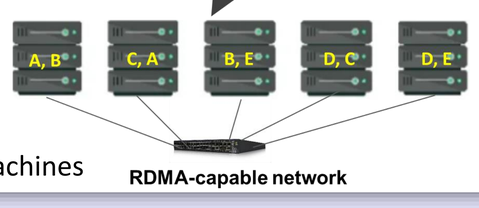
Dandelion: across-stack innovation + SW-HW co-design + batched execution (OLTP *like* OLAP)!

Dandelion

Context

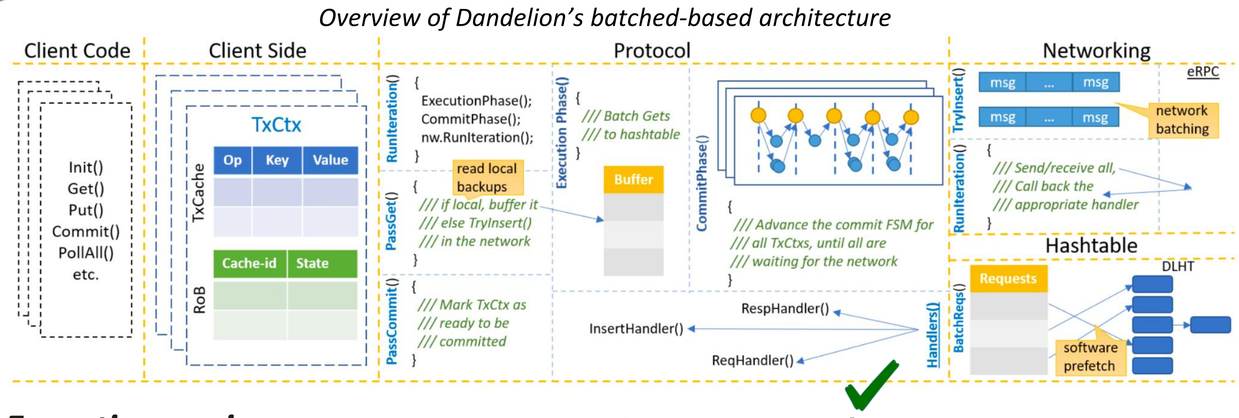
In-memory dataset: small/moderate sized **key-value** pairs
- **Distributed** across multiple machines for capacity
- **Replicated** for availability in the face of faults
- **Interactive Transactions**

```
tx.Init();
  auto val = tx.Get(A)
  if (val == 0) {
    tx.Put(B, 0);
  }
  tx.Insert(C, 1);
  tx.Delete (D)
tx.Commit();
```



Goal
1. **Reliable**: availability, strong consistency
2. **Programmable**: expressive, transactions
3. **Top performance/cost**:
e.g., better performance with 10x fewer machines

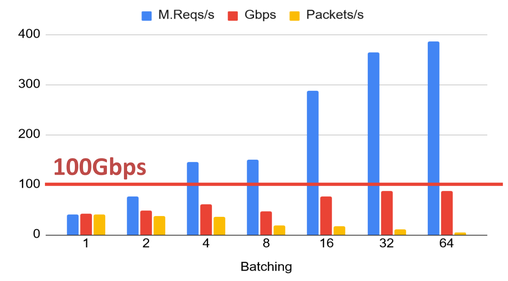
Overview of Dandelion's batched-based architecture



Execution engine: end-to-end **asynchronous & batched-based** → avoids wasting CPU cycles

Network

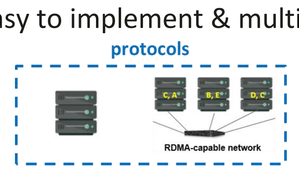
Datastore-optimized RPCs-over-RDMA ✓



Application-level batching
- necessary to fully utilize modern (100s Gbit net.)
- dandelion can batch 10s requests in one packet

Protocols

Easy to implement & multiplex protocols efficiently in Dandelion

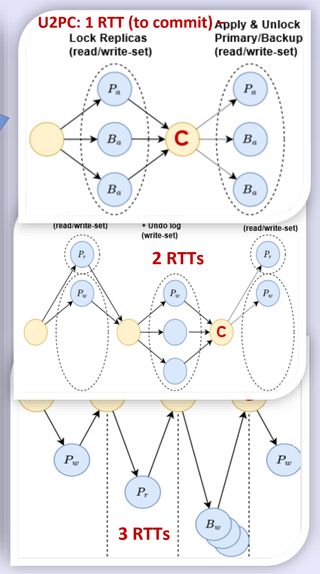


PaPoC'24
Unanimous 2PC: Fault-tolerant Distributed Transactions Can be Fast and Simple

3 Novel Fast Reliable Protocols: latency, throughput, lock-window
- Efficient logging and decentralized recovery
- Strongest consistency, Correctness: verified in TLA+
- High performance

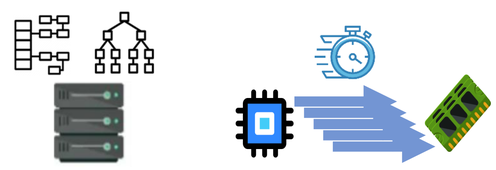
Lock-free execution phase (+ caching) ✓
- neither reads nor writes grab locks →
- programmability, performance, simple recovery

Fast reliable commit phase
- read-only / write-only txs: 0-1RTT
- read/write txs: 1RTT – 3RTTs (based on protocol) ✓



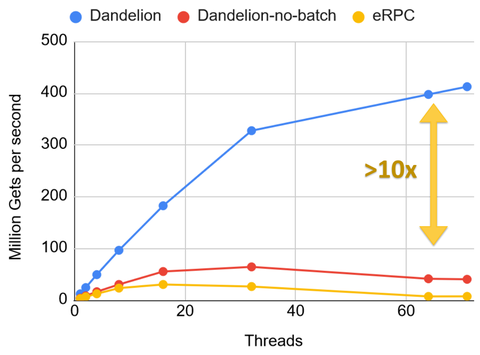
Data structures

Practically non-blocking and **memory-aware** ✓

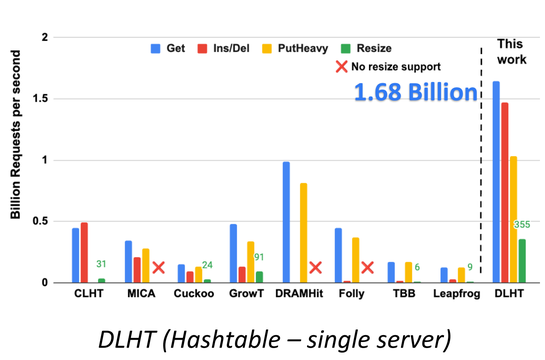


Dandelion Hashtable (DLHT) [Tree (DLTree)]
1. **Non-blocking requests [reads]**
2. **Most requests: 1 [few] memory access**
3. **Pipelined batched request processing**
→ overlap memory access + in-order completion
4. **Transaction-native and batched API** …

Networking

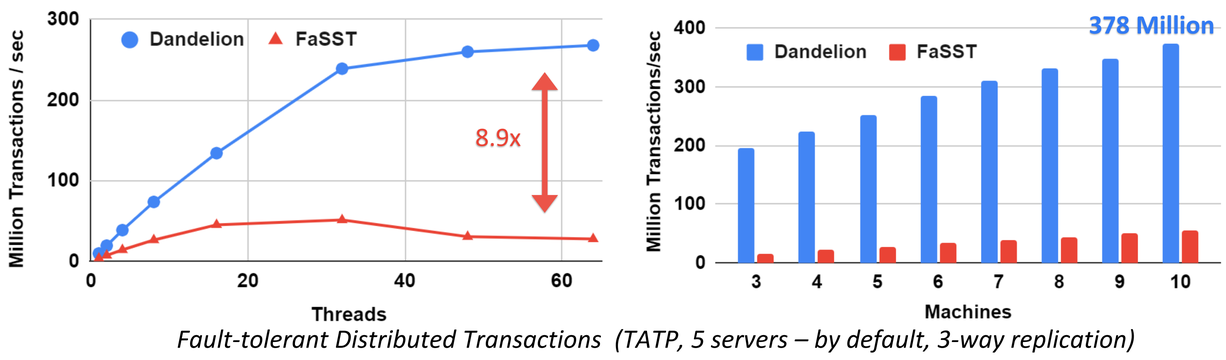


Data Structures



DLHT (Hashtable – single server)

Dandelion: network + protocols + data structures + async. batched engine



Fault-tolerant Distributed Transactions (TATP, 5 servers – by default, 3-way replication)

Dandelion networking **>10x** over eRPC

Dandelion Hashtable **>1.6B ops/sec** on a single server

Dandelion **>250M** distributed replicated txs (5 servers) | **12x-7x** faster than FaSST (3-10 servers)

Commodity servers: *18-core Intel Xeon Gold 6254 (2 sockets), 128GB DRAM* Network: *100Gbit RoCE*

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